

KHALKGOL, Mongolia

WMO: 443130

Lat: 47.6300N

Lon: 118.6222E

Elev: 2257

StdP: 13.54

Time Zone: 8.00 (E08)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.4	-30.4	-38.6	0.7	-35.1	-33.7	0.9	-30.1	26.0	0.2	21.2	-2.2	2.2	180	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	90.4	66.3	86.5	64.9	83.1	63.7	71.4	82.9	68.8	79.5	66.8	77.0	11.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	111.6	76.5	65.5	102.3	73.8	63.3	94.8	71.8	36.9	83.9	34.6	79.5	32.8	76.9	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	20.9	19.6	-41.5	96.5	4.8	4.5	-45.0	99.7	-47.8	102.3	-50.5	104.8	-54.0	108.1
			-41.5	73.7	5.1	3.6	-45.2	76.3	-48.2	78.4	-51.1	80.5	-54.8	83.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	-11.1	-3.1	15.1	37.8	52.7	64.7	70.0	66.2	53.6	35.8	12.2	-5.4	(6)
(7)		DBStd	30.09	10.30	11.38	13.20	9.43	8.16	7.02	5.32	6.13	8.13	9.36	12.12	10.25	(7)
(8)		HDD50	8261	1895	1487	1081	381	63	0	0	1	54	448	1134	1717	(8)
(9)		HDD65	12199	2360	1907	1545	817	391	88	13	56	352	904	1584	2182	(9)
(10)		CDD50	1896	0	0	0	15	147	442	620	502	161	9	0	0	(10)
(11)		CDD65	358	0	0	0	0	10	79	168	92	9	0	0	0	(11)
(12)		CDH74	4244	0	0	0	12	246	1104	1682	1011	186	3	0	0	(12)
(13)		CDH80	1467	0	0	0	1	73	408	608	332	45	0	0	0	(13)
(14)	Wind	WSAvg	7.3	5.7	6.1	7.3	9.3	9.4	7.8	7.1	7.2	7.8	7.7	6.5	5.8	(14)
(15)	Precipitation	PrecAvg	11.8	0.1	0.1	0.1	0.4	0.6	1.7	3.8	2.6	1.2	0.5	0.2	0.1	(15)
(16)		PrecMax	23.5	0.5	0.3	0.5	2.3	1.8	5.9	8.3	7.6	3.6	1.8	0.7	0.4	(16)
(17)		PrecMin	4.1	0.0	0.0	0.0	0.0	0.1	0.0	0.8	0.4	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.3	0.1	0.1	0.1	0.4	0.4	1.3	1.9	1.6	0.9	0.4	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	34.2	55.6	75.2	87.6	94.7	95.9	93.9	85.7	71.0	48.0	26.7	(19)
(20)			MCWB	16.7	27.9	41.8	51.8	59.1	65.1	71.9	69.8	61.4	51.7	37.8	23.5	(20)
(21)		2%	DB	13.9	27.3	46.0	66.6	81.0	88.8	90.9	87.4	79.6	62.7	40.4	18.4	(21)
(22)			MCWB	12.3	23.9	36.4	47.2	57.2	63.7	68.4	67.5	58.7	46.7	33.2	16.8	(22)
(23)		5%	DB	9.5	21.2	40.2	60.9	75.2	84.8	86.5	83.5	74.4	56.9	35.2	14.1	(23)
(24)			MCWB	8.4	18.8	32.4	45.0	54.2	62.6	66.0	65.0	56.5	44.6	30.3	12.8	(24)
(25)		10%	DB	4.9	15.8	34.9	55.3	69.9	80.5	82.8	79.7	69.6	51.9	30.2	10.3	(25)
(26)			MCWB	4.1	14.1	29.4	41.8	52.2	60.8	65.4	63.3	54.6	42.2	26.8	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	28.9	42.5	55.1	62.8	70.6	76.0	75.7	65.8	54.0	39.2	23.7	(27)
(28)			MCDWB	19.2	33.8	54.2	68.7	78.6	83.6	89.3	87.8	77.2	65.9	48.0	27.1	(28)
(29)		2%	WB	12.4	24.0	36.9	49.1	59.0	67.1	72.1	71.3	61.9	48.6	34.7	16.9	(29)
(30)			MCDWB	13.8	27.7	44.9	65.2	75.8	80.7	83.8	82.7	73.0	61.5	41.1	18.3	(30)
(31)		5%	WB	8.4	18.7	33.0	45.3	56.3	64.9	69.7	68.0	59.3	45.5	31.5	12.8	(31)
(32)			MCDWB	9.5	21.0	39.3	59.0	72.0	78.3	80.7	77.7	69.7	55.1	36.5	14.1	(32)
(33)		10%	WB	4.1	14.0	29.6	42.2	53.5	63.2	67.8	65.7	56.6	42.8	28.5	9.3	(33)
(34)			MCDWB	4.9	15.6	34.8	53.9	68.0	75.7	77.7	75.4	66.3	51.3	32.5	10.3	(34)
(35)	Mean Daily Temperature Range	MDBR	21.0	24.3	24.0	22.9	24.9	24.2	20.9	22.1	24.1	22.3	20.6	20.2	(35)	
(36)		5% DB	MCDBR	23.6	28.2	27.6	32.7	34.9	31.2	28.4	28.9	32.0	31.0	25.7	21.1	(36)
(37)			MCWBR	22.2	25.1	18.9	18.5	16.8	12.4	10.9	12.1	15.1	17.9	17.7	19.7	(37)
(38)		5% WB	MCDBR	23.3	27.6	26.4	30.5	30.2	25.1	22.6	23.2	25.9	28.5	24.6	21.1	(38)
(39)			MCWBR	22.1	24.8	18.7	18.3	16.0	11.9	11.3	11.9	14.0	17.0	18.0	19.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.285	0.335	0.407	0.485	0.480	0.437	0.402	0.356	0.325	0.292	0.270	(40)	
(41)		taud	2.229	2.178	2.122	2.043	1.875	2.013	2.211	2.279	2.340	2.327	2.262	2.199	(41)	
(42)		Ebn at Noon	259	277	277	266	249	251	261	266	269	259	243	237	(42)	
(43)		Edh at Noon	25	33	41	49	61	53	43	39	33	28	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	621	1021	1481	1732	1945	1975	1866	1721	1391	981	626	480	(44)	
(45)		RadStd	39	56	75	112	124	146	130	121	104	66	36	42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	-4.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page